Chapter 16

Introduction to Composite Resins

"Pleasure in the job puts perfection in the work".

—Aristotle

History

Prior to the introduction of composite resin materials, dentists had two types of tooth-coloured restorative materials that could be utilized to restore carious defects in anterior teeth: *silicate cement* and polymethyl methacrylate (*PMMA*).

I. **Silicate cement**
   - *Composition:* Silicate cements were supplied as a powder and liquid, with the main constituent of the powder being ground aluminosilicate glass and the liquid phosphoric acid.
   - These materials demonstrated decent initial esthetics, but had poor physical properties and colour stability and most importantly dissolved readily in oral fluids.
   - Silicate restorations were expected to serve 3–5 years, at which time they would have to be replaced. The main advantage of this material was that as it dissolved, it released substantial amounts of fluoride, with the result that recurrent caries almost never occurred with silicate restorations.

II. **Polymethyl methacrylate (PMMA)**
   - *Composition:* The second available tooth coloured direct material was PMMA that were also supplied as a powder and liquid. The powder was essentially ground prepolymerized methyl methacrylate and initiator, and the liquid was monomeric methyl methacrylate.
   - PMMA materials provided good initial esthetics but also had poor physical properties and lacked colour stability; however, their main disadvantage was considerable polymerization shrinkage (three times the shrinkage of modern composites) resulting in large contraction gaps between the restoration and the cavity margins. These gaps permitted significant microleakage that resulted in a very high incidence of recurrent caries. Thus, while PMMA materials were chronologically developed later than silicate materials, they were actually inferior in terms of clinical performance.

Composite Resin Materials

These materials were introduced to the profession in 1955 by *Dr Raphael Bowen*.[1] It is interesting to note that the discovery of the concept of acid-etching enamel by *Dr Michael Buonocore* occurred almost at the same time.[2]

The initial composite resin materials were superior to their predecessors (silicate and PMMA) in terms of physical properties, colour stability, solubility and clinical performance. However, these initial materials were not suitable for use in posterior teeth and had problems resulting from polymerization shrinkage even though the amount of that shrinkage was substantially less than the shrinkage of PMMA materials. Composite resin materials have evolved constantly over the past 50 years and contemporary composite resin materials are vastly superior to the original material in both clinical performance and esthetic potential.

Composition of Composite Resin Materials

There are three main components of all composite resin materials (Fig. 16.1). These are the:

I. *Resin matrix*:

   The resin matrix of most composite resin materials is **Bis-GMA** (Bisphenol-glycidyl methacrylate). Some composite resins utilize urethane dimethacrylate

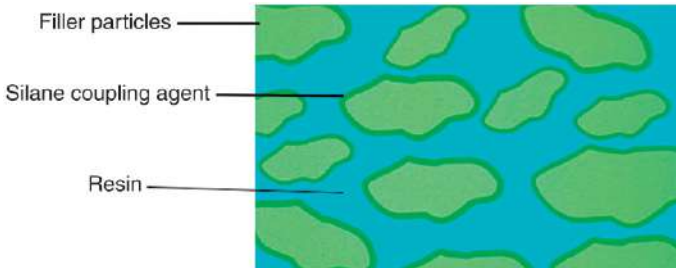


Figure 16.1 Main components of composite resins.

(UDM) as the matrix and both resins are substantially equivalent in terms of clinical performance.

Bis-GMA was synthesized from a type of epoxy by Bowen. It sets by means of a polymerization reaction and its major advantage over methyl methacrylate is that the monomeric molecule is substantially larger than monomeric methyl methacrylate. This results clinically in significantly less polymerization shrinkage.

II. *Filler particles*:

The filler particles in composite resin are composed of quartz or silica, and vary considerably in terms of size and shape (Fig. 16.2A and B). The incorporation of fillers into the resin matrix increases the strength of the material and further reduces the amount of polymerization shrinkage.

There are two rules regarding filler particles in composite resins:

- *The first rule is the more filler, the better*.
  - Increasing the relative amount of filler in a material increases the strength and reduces the amount of polymerization shrinkage.
  - Most modern composites experience a 2.4%–2.8% polymerization shrinkage.
  - A contemporary composite resin material for restorations should contain at least 75% filler by weight. Most modern materials meet or exceed this specification.
- *The second rule regarding filler particles is the smaller the average filler particle size, the better*.
  - Materials with small filler particles are inherently more polishable, retain their polish for longer periods and have improved wear resistance.

  One difficulty that manufacturers have faced is that it is difficult to maintain a high filler content with very small filler particles. This is because small filler particles possess a high ratio of surface area to weight, and this resulted in materials that were not usable with high filler content. Modern technology has resulted in a constant evolution of materials, the crux of which has been the ability to create useful materials with a high filler content using smaller and smaller particles.

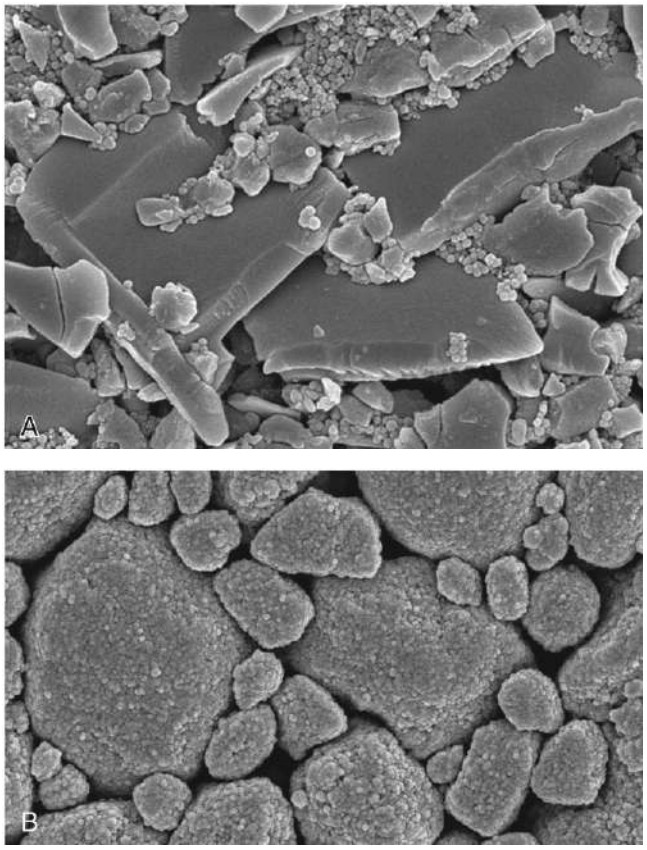


Figure 16.2 (A and B) SEM of composite resin displaying different sizes and shapes of filler particles.
(Images Courtesy: Dr Jorge Perdigao.)

III. *Silane coupling agents*:

- The filler particles are chemically bound to the resin matrix by silane coupling agents.
- Silanes are complex, bifunctional molecules that have two different end groups, one of which bonds to the filler particle and the other bonds to the matrix. The use of silane allows for stronger bonding of the filler to the matrix, which improves wear resistance, and also permits the incorporation of more filler into a given amount of resin matrix. The silanization process is complex and involves coating the particles with silane and then utilizing acetone washes to thin the thickness of the silane to a monomolecular layer.

Composite resin materials also contain numerous other components including:

- Pigments,
- Viscosity diluents,
- Cross-linking agents and
- Initiators: Most current composites are polymerized with the help of light. Hence, they employ *camphorquinone* as the photoinitiator, and it absorbs photons of light energy, predominantly at about *470 nm*.

Box 16.1 Classification of dental composites

I. **Based on polymerization method**
   i. Light cured composites
   ii. Chemical cure composites
   iii. Dual cure composites
II. **Based on filler particle type**
   i. Macrofilled composites
   ii. Microfiliied composites
   iii. Hybrid composites
      – Microhybrid
      – Nanohybrid
III. **Based on viscosity**
   i. Packable composites
   ii. Flowable composites
IV. **Recently introduced composites**
   i. Low shrinkage composite resin materials
   ii. Silorane composites
   iii. Bulk-fill composite resin materials
      – Flowable base bulk-fill composites
      – Full-body bulk-fill composites

Classification of Composite Resin Materials

Several different classifications for composite resin restorative materials have been proposed over the years, for example, according to polymerization reaction initiation or according to the size of filler particles. Refer to Box 16.1 for classification of contemporary composite resin restorative.

I. According to Polymerization Reaction Initiation

The polymerization reaction for composite resin materials can be initiated chemically or with visible light.

A. Chemical Cure

Initial composite resin materials were chemically initiated and were called *chemical-cure*, *auto-cure* or self-*cure* composite resins. These materials had a limited working time and suffered from poor long-term colour stability as they contained tertiary amines that turned yellow or orange after several years of service. Chemical cure materials are used today primarily as core materials for the buildup of extensively damaged teeth (Fig. 16.3).

B. Light Cure

The original light-cure materials were cured using ultraviolet light, but contemporary composite materials all utilize visible light, which will activate the initiator (camphorquinone). Light-cure materials have the substantial advantage of being able to cure on command and thus provide essentially infinite working time so that the restoration can be sculpted and shaped into final form prior to polymerization. They are also colour stable due to the elimination of the tertiary amines (Fig. 16.4).



Figure 16.3 Chemical cure composite resin.
(Courtesy: BISCO.)

C. Dual Cure

Dual-cure composite resin materials, which utilize both chemical cure and light cure technologies, are primarily used as cements or core materials. With these materials, the setting reaction is initiated by exposure to visible light but the reaction will slowly continue over time in the absence of light (Fig. 16.5).



Figure 16.4 Light cure composite resin.
(Courtesy: ESPE.)



Figure 16.5 Dual cure composite resin.
(Courtesy: BISCO.)

II. According to Filler Particle Size

Early composite resin materials utilized rather large filler particles and are called *'macrofilled'* composite resins. As esthetic composite bonding became popular, manufacturers developed composite resin materials with very small fillers that are called *'microfilled'* composite resins. Modern composite resin materials contain a variety of sizes of filler and are called *'hybrid'* composite resins. The evolutionary trend has been to develop hybrid composite resin materials with smaller and smaller fillers. Thus, contemporary hybrid composite resin materials may be subdivided into subgroups of *'microhybrid'* and *'nanohybrid'* composite resin materials. Other categories of contemporary composite resin materials include *'flowable'* composite resins and *'packable'* composite resins.

A. Macrofilled Composite Resin Materials

The original composite resin materials were macrofilled composite resins.

These materials had a relatively high level of loading (approximately **75%** by weight), with particles that ranged in size between **4 and 40 microns.**

Commercial examples of macrofilled composite resins include Adaptic (Johnson & Johnson, New Brunswick, NJ, USA) and Concise (3M Oral Care, St. Paul, MN, USA).

Because of the relatively high loading these materials were quite strong, but the rather large size of the filler particles resulted in two significant disadvantages.

- First, these materials were not polishable. The filler particles were harder than the abrasives used to polish them. As these materials were polished, the relatively soft matrix was worn away, thus exposing the large, hard fillers; and the more these materials were polished, the rougher they got. This is obviously a problem for an 'esthetic' restorative material (Fig. 16.6).
- Second, these large particles could be 'plucked' from the surface of a posterior restoration by opposing cusp tips and result instantly in a quantum loss of restorative material (40 microns). Thus the materials had very poor wear resistance, a major deficiency clearly demonstrated in early clinical trials (Fig. 16.7).

B. Microfilled Composite Resin Materials

The poor polishability of macrofilled composite resins directly led to the introduction of microfilled composite resin materials.

These materials had a relatively low level of filler loading (**45%–55%** by weight), with a uniform filler particle size of **0.04 microns.**



Figure 16.7 Maxillary premolars and first molar restored with macrofilled composite resin.

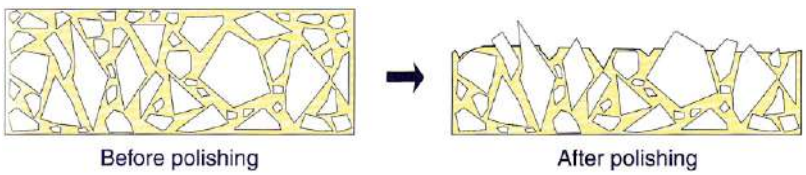


Figure 16.6 Schematic illustration of macrofilled composite resin before and after polishing.

This resulted in restorations that were very polishable and thus provided an excellent esthetic result. The low level of filler loading resulted in a material with a low modulus of elasticity (MOE) which made it unsuitable for stress bearing restorations.

These materials were contraindicated for Class IV restorations and for the restoration of posterior teeth. This low modulus of elasticity made these materials the material of choice for the restoration of NCCLs which many consider to have a flexural etiology. The low MOE allows the material to flex along with the tooth.

Commercial examples of microfilled composite resin materials include Silux Plus (3M Oral Care, St. Paul, MN, USA), Renamel (Cosmedent, Chicago, IL, USA), Durafill VS (Kulzer, South Bend, IN, USA) and A110 (3M Oral Care, St. Paul, MN, USA). Current evidence indicates that microfilled composite resins may not be needed because the filled bonding agents act as stress breakers.

C. Hybrid Composite Resin Materials

Hybrid composite resin materials have a combination of small and large filler particles, to combine high filler loading with a sufficient quantity of small particles to ensure improved wear resistance and also permit adequate polishability. These materials have been in use since the early 1980s and have proven to be excellent materials for both anterior and posterior restorations.

Most contemporary hybrids have a filler content of **75%–80%** by weight, and the trend has been to produce materials with smaller and smaller average filler particle size. With most contemporary materials the largest filler particle would be in the range *between 1 and 2 microns*. These materials are quite strong, polishable and have wear resistance equivalent to amalgam (6–15 microns/year). The esthetic potential of these materials is excellent and many products offer a wide range of shades and translucencies. Typical commercial products in the hybrid category include Herculite XRV (Kerr Corporation, Orange, CA, USA), Z100 (3M Oral Care, St. Paul, MN, USA) and TPH (Dentsply Sirona, York, PA, USA).

The most contemporary hybrid composite resin materials have been described as ***microhybrid***s and ***nanohybrids***. These materials contain a mixture of small and smaller particles and have excellent handling characteristics and esthetic potential. High filler loading is made possible with advanced technology that permits the formation of clusters of very small particles that reduce surface area yet function as individual nanoparticles to ensure adequate wear resistance. Commercial products in this category include Esthet-X (Dentsply Sirona, York, PA, USA), Point 4 (Kerr Corporation, Orange, CA, USA) and Filtek Supreme (3M Oral Care, St. Paul, MN, USA) (Fig. 16.8A–D). The major advantage of nanocomposites is that they polish to a very high luster and they maintain that luster over time.

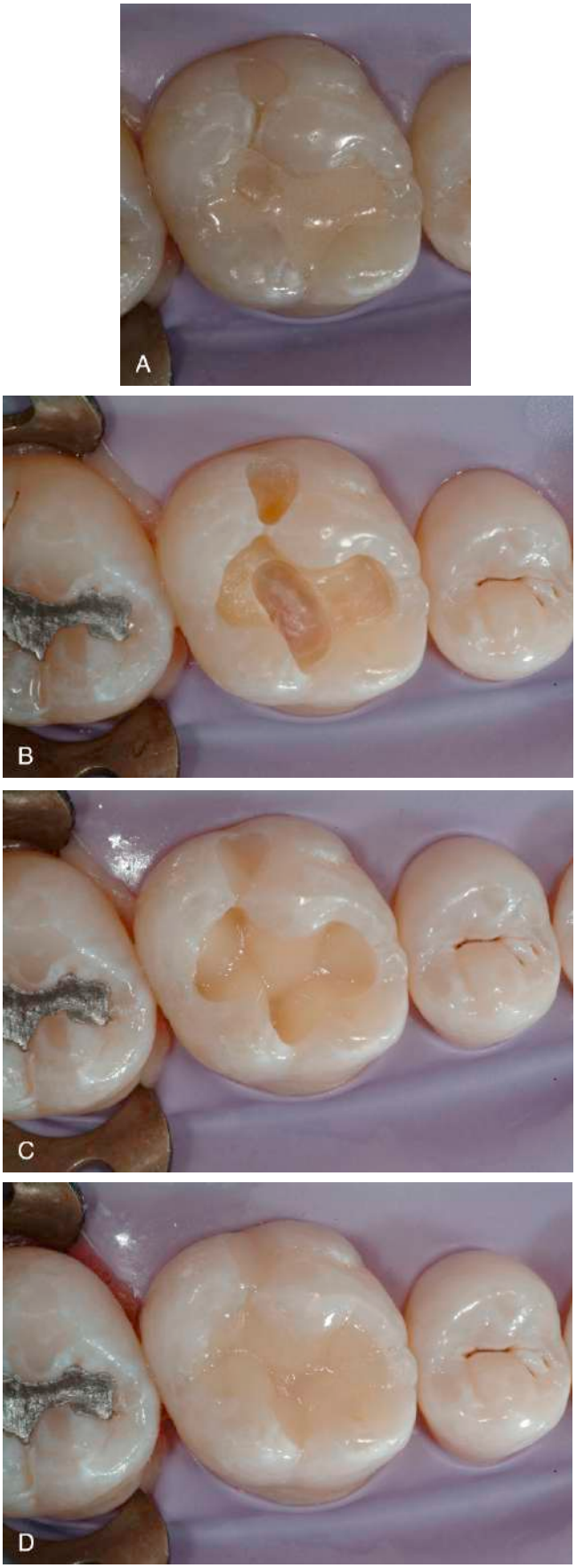


Figure 16.8 (A–D) Maxillary first molar restored with a hybrid composite resin.
(Photos Courtesy: Dr Phillip Timmins.)

III. According to Viscosity

A. Packable (Condensable) Composite Resin Materials

Several products have been introduced to the market in recent years that have been described as 'packable' or 'condensable' composite resins. These materials have been designed by manufacturers to possess handling characteristics similar to amalgam.

These materials are basically *hybrid composite resins to which large fillers have been added that can be condensed or packed* in a manner similar to amalgam.

The addition of these large fillers has resulted in documented *poor wear resistance* of some of these materials. While some dentists prefer the handling characteristics of these materials, it is doubtful that the benefits of improved handling offset the reduction in clinical performance that has been demonstrated by some of these products.

B. Flowable Composite Resin Materials

In recent years manufacturers have introduced a group of materials designated as 'flowable' composite resin materials. These materials represent a wide range of products, with filler content ranging between **35% and 65%** by weight. These materials are very convenient to use because their viscosity allows them to be injected into the cavity preparation with a syringe. However, as would be expected, the reduction in filler content is accompanied by reduced physical properties and increased polymerization shrinkage. The use of these materials should be restricted to 'niche' functions such as a *0.5-mm-thick liner under large posterior composite restorations*.

IV. Recently introduced composites

A. Low-shrinkage Composite Resin Materials

Most modern hybrid composite resin materials exhibit linear polymerization shrinkage of between **2.2% and 2.4%**. This amount of shrinkage, if not compensated for by clinical technique, can result in contraction gap formation, microleakage, postoperative sensitivity and a potential to develop recurrent caries.

Some manufacturers have introduced products to the market that are described as 'low shrinkage' composite resin materials. The linear shrinkage of these materials ranges between **1.4% and 1.7%.** The reduction in shrinkage has been accomplished by technology that permits higher filler loading. These materials have not undergone extensive clinical testing at the time of writing. More clinical evidence of their efficacy is essential before they can be recommended for routine use.

B. Silorane Composites

Recently, composite resin materials based on **silorane technology** have been introduced on an experimental basis. These materials use a resin polymer that exhibits very low shrinkage and thus does not depend on increased filler loading.

As with other low shrinkage materials, more clinical evidence is required before these materials can be considered for routine use.

C. Bulk-fill Composite Resin Materials

Bulk-fill resin materials were recently introduced to allow clinicians to divert from using an incremental layering technique when placing composite restorations, thus simplifying the procedure and reducing chair time. This technique was developed to minimize negative effects of polymerization-induced shrinkage stress and curing light penetration. The undesired side effects—a time-consuming and complex clinical procedure—contribute significantly to make composite restorations technique sensitive.[3]

Different from traditional composites, which are typically restricted to increments of 2 mm or less, these materials are designed to be used in *increments of 4–5 mm*.[4,5] Common manufacturers' claims involve greater depth of cure and lower polymerization-induced shrinkage stress. Although these composites are usually referred to as if they were clearly identifiable as a class of materials—bulk-fill composites—scientific evidence contradicts the assumptions that they would behave similarly.[6,7] The variety in handling characteristics, as well as physical and mechanical properties, is a direct consequence of the wide range of different compositions used by manufacturers, largely related to their filler content.[8]

The different types of bulk-fill composites currently available can be categorized as:

I. *Flowable base bulk-fill composites*: Flowable base bulk-fill composites require a conventional composite as the occlusal increment and therefore are used for dentin replacement only.
II. *Full-body bulk-fill composites*: Full-body bulk-fill composites can replace dentin and enamel in a single increment.[7]

Overall, bulk-fill composites are more translucent than other composite resin materials. To a certain extent, this is attained by decreasing the filler content and increasing the filler size. Increased translucency facilitates light penetration, which is closely associated with depth of cure and increased polymerization rates towards the deepest areas of the restoration.[9] Evidence appears to support light-curing times of at least 20 s.[7]

Despite the large array of materials being marketed, bulk-fill composites appear to deliver an increased depth of cure and reduced polymerization-induced shrinkage stress.[10,11]

Manipulative Variables with Composite Resin Restorative Materials

The contemporary clinician has a wide variety of quality composite resin restorative materials from which to choose. *How* these chosen materials are manipulated in the clinical setting is more important to long-term results than *which* material is chosen. The following narrative will briefly discuss some of the important variables when using composite resin restorative materials.

- Most contemporary composite resin materials are cured or set by means of exposure to visible light using a curing light. In general, most composite resin materials should not be cured in thick sections, and the *recommended thickness of an increment of composite resin is about 2 mm*. Curing sections greater than 2 mm may result in insufficient curing of the undersurface of the composite resin material. The depth of cure is determined by a number of variables:
  - The shade of the composite resin.
  - The thickness of the material.
  - The irradiance of the curing light and the time of cure.
- Curing lights should be checked periodically with a radiometer (most curing lights have one built in) to ensure that the light emitted is of sufficient output irradiance to efficiently cure the material. The light should be held in close proximity to the material (the closer the light, the greater the irradiance) and the composite resin should be exposed to the light for a sufficient time frame. In this regard, the manufacturers' recommended curing times should be closely followed.
- The major deficiency of composite resins is their inherent polymerization shrinkage (2.2%–2.4%). This can result in the formation of substantial contraction gaps between the restorative material and the cavity margin, which can result in postoperative sensitivity and recurrent caries. The shrinkage of composite resin materials cannot be avoided, but the stress created by the shrinkage can be reduced by the placement technique. Reducing polymerization shrinkage stress can result in intact restoration margins and the prevention of microleakage.

There are ***five primary strategies*** that can be used to reduce polymerization shrinkage stress:

I. The first is to create a *relatively thick primed layer* with the chosen dentin bonding agent. OptiBond FL (Kerr Corporation, Orange, CA, USA) has a relatively high amount of filler content and, when properly used, will result in a thick primed layer. This primed layer has a relatively low modulus of elasticity and can thus deform slightly when the composite resin is curing and shrinking and absorb some of the stress and maintain the integrity of the tooth/restoration bond.

II. The second strategy is to use a *thin liner* (*0.5 mm*) *of resin-modified glass ionomer* (Vitrebond Plus, 3M Oral Care, St. Paul, MN) under the composite resin restoration. These materials bond predictably to tooth structure and also have a low modulus of elasticity to absorb the shrinkage stress.

III. The third option is to place a *thin layer* (*0.5 mm*) *of flowable composite resin* as a liner. This again will theoretically effectively absorb the shrinkage stress.

IV. A fourth concept that may be used to reduce shrinkage stress is the use of *'soft-start' polymerization*.[12] This approach utilizes a number of procedures to prolong the gel phase of the setting reaction of the composite resin material. The theory is that the longer the setting composite can maintain a gel or flowable condition, the better it can distribute the stresses resulting from shrinking. Procedures to do this include the use of ramped lights, pulse curing or simply holding the curing light some distance from the material for the initial exposure. Several clinical and laboratory studies have demonstrated the effectiveness of soft-start polymerization, while other studies have demonstrated limited benefit.

V. *C-factor*: The final approach to reduce stress is to *place the composite resin in increments, always keeping in mind the concept of 'C-factor'*.[13] The C-factor relates to the configuration of the cavity walls, and *is the relative ratio between the bonded surface area of the composite resin to the unbonded surface area*.
   - The higher the C-factor, the greater is the potential for bond disruption from polymerization effects.
   - A Class IV composite restoration has a wealth of unbonded surface area and a relatively small bonded area and thus has a low C-factor. The unbonded composite resin material will shrink towards the bond, thus reducing the shrinkage stress (Fig. 16.9).
   - On the other hand, small Class I cavities have a high C-factor because they have a huge bonded area and relatively small unbonded area. When these restorations are cured, there is a high amount of competition

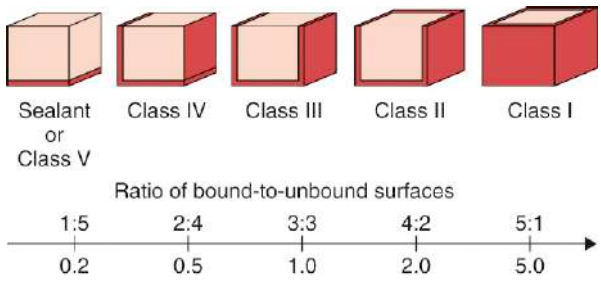


Figure 16.9 Configuration factors (C-factors) associated with polymerization shrinkage for different situations using dental restorative materials. C-factors are the ratio of bound-to-unbound surface areas on restorations and are shown in the figure as calculated using average estimates of dimensions of tooth preparations.
(From Feilzer AJ, et al.: Setting stress in composite resin in relation to configuration of the restoration, *J Dent Res* 66:1636–1639, 1987).

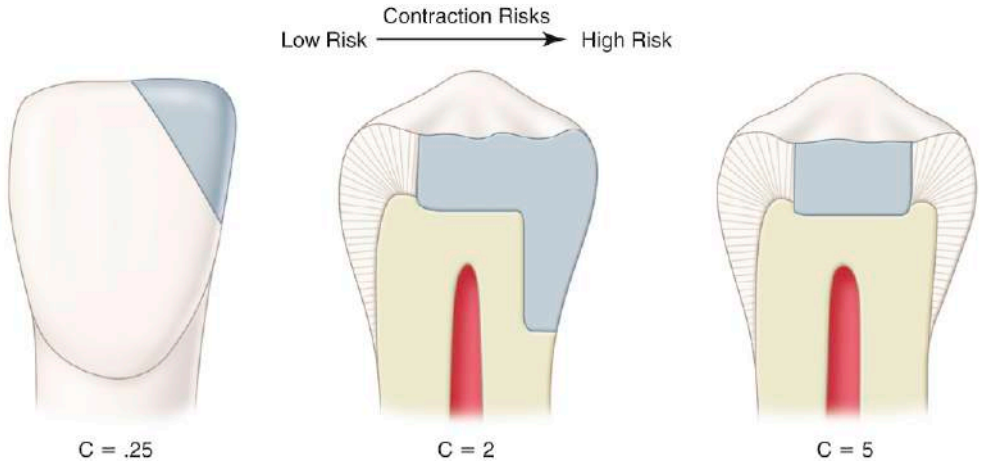


Figure 16.10 Configuration factor (C-factor) (bonded surfaces/unbonded surfaces).

for the bond, and the result is that the weakest bond (pulpal floor) gives away under the shrinkage stress creating a fluid-filled gap below the restoration and a high likelihood of postoperative sensitivity. These restorations should be placed in anatomical increments so that the material can shrink freely in one direction and thus eliminate or reduce the shrinkage stress (Fig. 16.10).

Summary

Contemporary composite resin materials have improved significantly over the years. Today's materials have good wear resistance and a wide variety of shades and translucencies. They are quite polishable and provide excellent esthetics.

However, the major disadvantage of composite resin materials is the inherent polymerization shrinkage. This results in a significant level of technique sensitivity. Composite resin restorations must be placed with excellent isolation (rubber dam), in conjunction with meticulous bonding procedures. The placement technique itself is critical, as is meticulous finishing and polishing. Properly placed, composite resin materials can provide a very esthetic and durable restoration, and the results are gratifying to both patients and clinicians.

Clinical Notes

- There is a significant interest by the public in tooth coloured restorative materials.
- Composite resin restorative materials have improved significantly over the years.
- High filler content (75% weight) of small filler particles (0.04–1 micron) is the key to that improvement.
- Polymerization shrinkage (2.4%–2.8%) is the main disadvantage of composite resin materials. This can be accounted for with proper technique.
- Composite resins must be used in conjunction with contemporary bonding systems.
- Excellent isolation is critical to success with composite resin materials.

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